

Guest Lecture: The Laws of Thermodynamics

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1 Introduction: Energy and Action

Basically, we need energy to perform any action. Have you ever imagined a car that doesn't need fuel or a battery? How would it start? What about a stopped object coming back to motion spontaneously?

To understand why these scenarios are impossible, we must look at **Perpetual Motion Machines**:

- **1st Kind**: Machines that produce work without any energy input.
- **2nd Kind**: Machines that spontaneously convert all thermal energy into useful work with 100% efficiency.

Unfortunately, the universe does not allow any of these to exist.

2 The Laws of Thermodynamics

2.1 0th Law

The 0th Law basically states that you can have a **thermometer**. If two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other.

2.2 1st Law: Conservation of Energy

This law defines **Work** and **Heat** as processes of transferring energy between systems.

- **Work (W)**: The transfer of energy mechanically.
- **Heat (Q)**: The transfer of energy thermally.

The change in the internal energy (ΔE) of a system is the sum of heat and work added to the system:

$$\Delta E = W + Q$$

Energy is conserved. You can do whatever you want as long as the energy you gain comes from another system. This rules out Perpetual Motion Machines of the **1st Kind**.

2.3 2nd Law: The Limit of Usefulness

Work is the "useful" energy transfer mechanism. While you can convert work into heat (e.g., friction), it is much harder to convert heat into work. The 2nd Law states that you cannot convert *all* heat (Q_H) to useful work (W). The efficiency (η) is always:

$$\eta = \frac{W}{Q_H} < 1$$

This rules out Perpetual Motion Machines of the **2nd Kind**.

3 The Kelvin Statement and Information

The Kelvin statement of the 2nd Law rules out the 2nd kind of perpetual motion. If you admit that heat always flows from a hotter body to a colder body, you are witnessing the equivalence of these statements.

The 2nd Law puts a severe restriction on what is possible. It arises from heat engines but tells us how much information we have about the state of the universe. When all of the universe reaches thermal equilibrium, we reach the **”Heat Death of the Universe.”**

4 Exercise: Information and Probability

Consider the following three statements regarding **Isaac Newton’s Birthday**:

1. Newton’s birthday falls on one of the 365 days of the year.
2. Newton’s birthday falls in the month of December.
3. Newton’s birthday is on December 25.

Discussion Questions:

- Which statement gives you the most **relevant information**?
- Which statement is **redundant**?
- What is the probability of these statements being correct assuming they are guesses proposed by three individuals who don’t know nothing about Newton at all?